When Fewer Steps Mean Lower Temperature: Euler Sampling of Flow-Matching Driving Policies

Anonymous Authors[1]

Abstract

Flow-matching driving policies are made faster by integrating with fewer Euler steps, and the change is usually reported as preserving or improving accuracy. We show that, at the level of the distributions they produce, few-step Euler samplers behave like an accurate reference solver run at a lower sampling temperature. One Euler step returns the conditional mean, and more steps contract samples toward it by an amount that a closed-form Gaussian factor matches at the median. On 60 clips from the PhysicalAI-AV test split, the released 10-step samplers of three Alpamayo generations share an effective temperature of 0.83–0.84, and the reference solver at that temperature reproduces 79–121% of their change in single-sample error and in the rate of changed driving decisions, while second-order Heun samplers with a similar budget fail the same test. The match holds for rates over clips, not for individual samples, and any contraction toward the mean of the same size would pass it; temperature is the explicit knob that reproduces it. We therefore propose that accuracy gains from fewer steps be reported against a reference solver at matched effective temperature, and we give a protocol and its statistics for doing so.

1. Introduction

Integrating Alpamayo-1.5 with four Euler steps instead of the released ten lowers its single-sample error against the human trajectory and changes about one decision in ten, compared with an accurate reference solver (Figure 1 and Table 1). Should the improvement be credited to the faster sampler? We argue that it should not: the same improvement, and nearly the same decision changes, are obtained by keeping the reference solver and shrinking the initial noise.

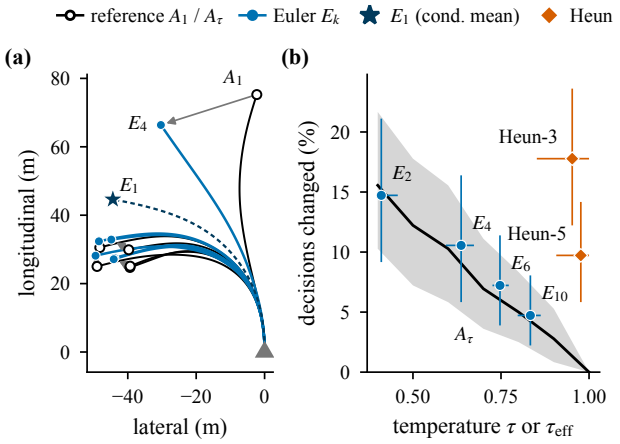


Figure 1. **Fewer Euler steps behave like a lower temperature; Heun steps do not.** (a) One PhysicalAI-AV test clip (Alpamayo-1.5, ego at 10.5 m/s), bird's-eye view, six shared noises. Four Euler steps (E_4, blue) move every sample of the reference solver (A_1, black) toward the one-step conditional mean (E_1, star); the one seed that goes straight under A_1 turns left. (b) Share of decisions that differ from A_1 under the same noise. Black: the reference solver started from τz (95% clip-bootstrap band). Euler samplers placed at their effective temperature fall on the curve; Heun-3 and Heun-5 change 17.8% and 9.7% of decisions at almost no change in temperature. 60 clips × 6 seeds; bars are 95% intervals.

The reason is elementary. Under rectified-flow interpolation (Lipman et al., 2023; Liu et al., 2023) the initial velocity is $v(z, 0) = \mathbb{E}[x_1 \mid c] - z$, so a single Euler step returns the conditional mean, and more steps move toward the exact flow. A k-step Euler sampler therefore sits between the mean and the policy's distribution, the same place a lower sampling temperature puts it. Accelerations that build on first-order integration (Li et al., 2026; Fang et al., 2026; Han et al., 2026; Tang et al., 2026) inherit this temperature shift, and averaged metrics such as minADE register it as an accuracy gain.

Contributions.

- **A matched-temperature evaluation protocol** for ac-

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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celerated flow samplers: estimate the sampler's effective temperature τ_{eff} from its seed spread, compare it with the reference solver run at τ_{eff}, and report effect, residual and equivalence with clip-level intervals (Section 4).

- **Evidence on three generations of a production driving VLA** on real test data. The released samplers share $\tau_{\text{eff}} \approx 0.83$; temperature reproduces 79–121% of the ADE_1 and flip-rate changes of E_4 and E_{10}; Heun-3 and Heun-5 fail the same test (Figure 3 and Table 1).

- **A mechanism with a closed form.** Contraction toward the conditional mean (Figure 2), the Gaussian factor $\prod_i \cos\theta_i$ that matches it at the median, and a symmetry condition under which no integrator changes a decision (Section 3).

- **A boundary.** Qwen-Drive's planners are nearly deterministic; there the solver adds spread rather than removing it, and the temperature view does not apply.

TODO: Add FlashDrive (official checkpoint and cache, matched temperature) and AlpaSim closed-loop contributions when the runs finish; keep only what the data support.

2. Background and Notation

Flow-matching driving policies. A VLA driving policy encodes cameras, ego history and a reasoning trace into a context c and decodes an action x_1 (64 normalized acceleration and curvature commands over 6.4 s in Alpamayo) by integrating $v_\theta(x, t \mid c)$ from $x_0 = z \sim \mathcal{N}(0, I)$ to $t = 1$. With $x_t = (1-t)z + tx_1$, the flow-matching minimizer is $v(x,t) = \mathbb{E}[x_1 - z \mid x_t = x, c]$. The released policies use forward Euler with 10 uniform steps (NVIDIA, 2025; Qwen Team, 2026).

Notation. E_k is Euler with k uniform steps; Heun-k uses $2k$ field evaluations. A_τ is the reference solver, Heun with 32 steps (64 evaluations), started from τz; A_1 is the reference at $\tau = 1$. All samplers for a clip share the same context, reasoning trace and six noises. $\text{Spr}(S)$ is the mean pairwise distance between the trajectories S produces from different noises. A *flip* is a change of decision class relative to A_1 under the same noise. ADE_1 is the mean single-sample displacement to the logged human trajectory.

Accelerations. FlashDrive reuses the velocity on steps 3–6 of an 8-step Euler schedule (Li et al., 2026); ProbeFlow and AdaVLA adapt the number of steps (Fang et al., 2026; Han et al., 2026); entropic time places steps by information about the output (Stančević et al., 2025); Coda corrects a few-step endpoint (Tang et al., 2026).

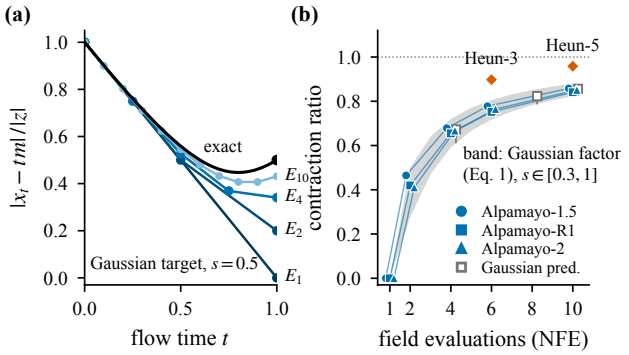


Figure 2. **Euler contracts samples toward the conditional mean by the Gaussian factor of Equation (1).** (a) One coordinate of a Gaussian target ($s = 0.5$): deviation from the mean path under the exact flow (black) and under k Euler steps (blue, darker = fewer steps). Each step loses a factor $\cos\theta_i$, so E_k ends at 0, 0.40, 0.68 and 0.86 of the exact deviation for $k = 1, 2, 4, 10$. (b) Median ratio of each sample's distance to E_1 over the reference sample's, same noise, against field evaluations (NFE); three Alpamayo generations, 60 clips × 6 seeds. Band: the Gaussian factor for $s \in [0.3, 1]$; squares: its prediction from Alpamayo-1.5's conditional action variance (median and IQR; a separate sample of 200 clips, action space). Heun samplers (vermillion) barely contract at the same budget.

3. Why Euler Steps Act Like Temperature

One Euler step returns the conditional mean. Since $x_0 = z$, $E_1(z) = z + v(z, 0) = \mathbb{E}[x_1 \mid c]$ for every z: the one-step sampler has no diversity at all.

Gaussian targets: contraction on any grid. For $p(x_1 \mid c) = \mathcal{N}(m, \text{diag}(s^2))$ the field is linear and coordinate d evolves with scale $\|u_d(t)\|$, where $u_d(t) = (1 - t, ts_d)$. One Euler step from t_i to t_{i+1} multiplies coordinate d's deviation from m_d by $\langle u_d(t_i), u_d(t_{i+1})\rangle / \|u_d(t_i)\|^2$, whereas the exact flow multiplies it by $\|u_d(t_{i+1})\|/\|u_d(t_i)\|$. Writing $\theta_{d,i}$ for the angle between $u_d(t_i)$ and $u_d(t_{i+1})$,

$$\frac{E_k(z)_d - m_d}{A_1(z)_d - m_d} = \prod_{i=0}^{k-1} \cos\theta_{d,i} \;\le\; 1 \qquad (1)$$

on any grid (Figure 2a). The inequality is implicit in the proof of Proposition 2 of Claustre et al. (2026); we use the closed form as a prediction. For $s_d \ge 0.3$ the factor of E_{10} lies between 0.82 and 0.88. Coordinates with smaller s_d contract more but carry little of the spread, so the aggregate factor is set by the high-variance coordinates.

Decisions protected by symmetry do not flip.

Proposition 3.1. *Let g be an orthogonal reflection with fixed hyperplane H, let $p(x_1 \mid c)$ be g-invariant, and let the decision be the side of H. An explicit Runge–Kutta integrator whose steps satisfy $h\,\text{Lip}(v(\cdot, t)) < 1$ returns the decision of the exact flow for almost every z.*

The field is g-equivariant and hence tangent to H on H; each step is an equivariant homeomorphism mapping H to itself and depending continuously on h, so half-spaces are preserved. Decision flips therefore need an asymmetric context.

TODO: full proof in the appendix; the Lipschitz condition can fail near $t = 1$, discuss.

4. Matched-Temperature Protocol and Setup

Definition 4.1. The effective temperature of a sampler S is the τ at which $\mathrm{Spr}(A_\tau) = \mathrm{Spr}(S)$, interpolated over $\tau \in \{1, 0.9, \ldots, 0.4\}$.

For a metric M we report $M(A_1)$, $M(S)$ and $M(A_{\tau_{\text{eff}}(S)})$. The *residual* $M(S) - M(A_{\tau_{\text{eff}}(S)})$ is the part of the sampler's effect that temperature does not explain, and $1 -$ residual/effect is the fraction explained. Clips are the independent unit: every quantity, including τ_{eff}, is recomputed in each of 1000 bootstrap resamples of clips, drawn jointly across models because all models see the same clips. We call S *equivalent* to $A_{\tau_{\text{eff}}}$ on a metric when the 95% interval of the residual lies inside a margin declared in advance (2 points for flip rates, 0.05 m for ADE_1). Fractions explained above 100% mean that the explicit temperature overshoots the sampler's effect; we therefore read residuals and their intervals, not fractions, as evidence. The protocol compares distributions: the same noise can still produce different decisions under S and $A_{\tau_{\text{eff}}}$ (Section 7).

Budgets. The reference solver uses 64 field evaluations (NFE). E_k uses k; Heun-3 uses 6 and Heun-5 uses 10, so Heun-5 has the released sampler's budget and Heun-3 sits between E_4 and E_{10}.

Policies. Alpamayo-1.5, Alpamayo-R1 and Alpamayo-2-Super, frozen; Qwen-Drive-1.0-4B (SFT and RL planners) as a boundary case. Our integrators run inside each model's own sampling call with the model's step function, so the reasoning trace is shared across samplers within a clip.

Data. The same 60 clips from the PhysicalAI-AV test split (NVIDIA, 2026) for every model (one clip per dataset chunk, sampled before any analysis), six noise seeds each, with the current frame at 5.1 s.

Decisions. Longitudinal (stop, decelerate, keep, accelerate) and lateral (left, right, lane shift, straight) classes are read from each plan with thresholds fixed in advance (Appendix E).

5. Results

Finding 1. *Euler samples contract toward the conditional mean; Heun samples do not.* In trajectory space, 92–93% of E_4 and 90–93% of E_{10} samples lie closer to the one-step mean than the reference sample from the same noise, with median ratios 0.66–0.68 for E_4 and 0.84–0.86 for E_{10} across the three models. The Gaussian factor of Equation (1), computed from Alpamayo-1.5's conditional action variance, gives 0.67 and 0.86 (Figure 2; per-model values in Table 2). It matches the median but not the ranking of clips, so we use it as an explanation of the typical value rather than as a per-clip predictor.

Finding 2. *The released samplers share one effective temperature, and temperature accounts for most of what they change.* $\tau_{\text{eff}}(E_{10})$ is 0.83 [0.80, 0.86], 0.84 [0.82, 0.85] and 0.83 [0.81, 0.85] for the three generations. The reference solver at τ_{eff} reproduces 79–121% of the ADE_1 and flip-rate changes of E_4 and E_{10} (Figure 3 and Table 1). Flip residuals are at most 2.0 points in absolute value. With 60 clips their intervals are too wide to establish equivalence at the 2-point margin in most cells, and one interval excludes zero: Alpamayo-2's E_{10} changes 2.0 points more decisions than its matched temperature [0.03, 4.4]. Euler is not temperature in every respect. Its best-of-six error (minADE_6) is worse than the reference's, as fewer distinct samples imply, and slightly worse than the matched temperature (residual 0.00 to 0.11 m, positive in 5 of 6 cells; Appendix A); decision-entropy residuals lie between -0.03 and 0.03.

TODO: pre-registered confirmation on fresh clips and seeds (the 436 clips of Appendix C were already used), primary endpoint E_{10} equivalence averaged over models.

Finding 3. *Second-order samplers fail the same test.* Heun-3 and Heun-5 barely contract (τ_{eff} 0.95 and 0.98 for Alpamayo-1.5), yet they flip 17.8% and 9.7% of decisions, against 1.3% and 0.6% for the reference solver at the same temperature; their flip residuals are $+16.5$ and $+9.1$ points (Figure 3 and Table 1). Heun-5 has the released sampler's budget. The test therefore discriminates: landing on the temperature curve is not a property of every cheap sampler. Part of Heun's residual is its own integration error rather than a departure from a distribution, and we do not claim that every non-Euler sampler fails.

TODO: Heun-3/5 on Alpamayo-R1 and -2; a Heun control matched in displacement from A_1.

Finding 4. *When the flow is nearly deterministic, the solver adds spread instead.* Under the reference solver, Qwen-Drive's planners have a seed spread of 0.04 m that τ barely changes, for both the SFT and the RL checkpoint. The released 10-step sampler has *more* spread (0.16 m) than the reference, and its best-of-six error is lower (1.42 vs. 1.56 m for the RL planner).

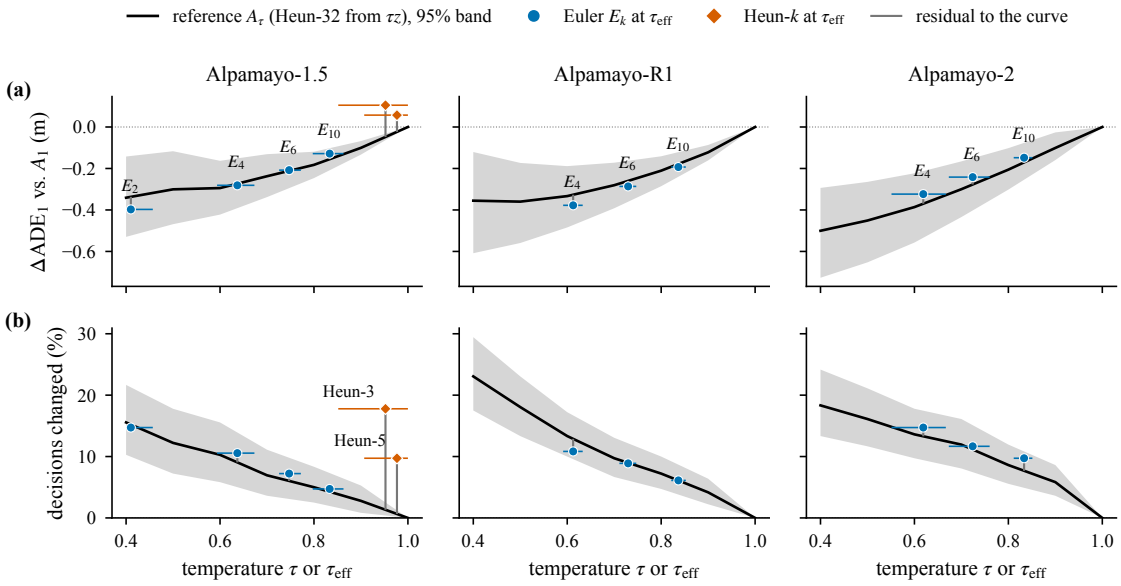


Figure 3. **Placed at its effective temperature, each Euler sampler lands on the explicit-temperature curve; Heun samplers do not.** Black: the reference solver started from τz, as a change from τ=1, with a 95% clip-bootstrap band. Blue: Euler with k steps at its effective temperature (Section 4; horizontal bars are 95% intervals); vermillion: Heun-3 and Heun-5. Gray ticks join each sampler to the curve at the same τ; their length is the residual of Table 1. (a) Single-sample error falls with temperature, and Euler follows it. (b) Euler changes about as many decisions as the reference at the same temperature; Heun changes 10–18% of decisions at $\tau_{\text{eff}} \approx 1$. E_1, and E_2 for Alpamayo-R1 and -2, have $\tau_{\text{eff}} < 0.4$ and are not shown. 60 PhysicalAI-AV test clips $\times$ 6 seeds per model. Best-of-six error is in Appendix A.

TODO: unify Qwen-Drive numbers with the 60-clip sample; check on its own evaluation protocol.

The one-step limit. E_1 contracts beyond the measured temperature range ($\tau_{\text{eff}} < 0.4$). At $\tau = 0.4$ the reference still changes 11–15 points fewer decisions than E_1, so our data do not show that E_1 is matched by any temperature.

TODO: reference at $\tau \in \{0.3, 0.2, 0.1\}$.

Rare decisions. Stopping is too rare in this sample to test: across the three models, the reference stops in 6, 6 and 6 of 360 plans, coming from 4 clips in which the vehicle is nearly stationary. We observe that lowering τ adds stops while fewer Euler steps remove them, and leave a test of this tail behavior to a sample enriched for stopping scenes. Log-replay collisions with logged agents are similarly rare (all differences between samplers come from 5 distinct clips) and are not used as evidence.

TODO: Finding on published accelerations: FlashDrive official checkpoint and cache at matched temperature (job running); minADE_k crossover on the unified sample.

6. Related Work

Accelerating VLA sampling. FlashDrive, ProbeFlow, AdaVLA and Coda reduce or correct flow sampling and report trajectory or success metrics (Li et al., 2026; Fang et al., 2026; Han et al., 2026; Tang et al., 2026). Our protocol separates the part of such gains that a temperature change reproduces.

Discretization and diversity. Few-step generators lose rare modes in image synthesis (Lin et al., 2026), few-step error has been tied to single-Gaussian assumptions (Chen et al., 2025), and Euler on Gaussian flows has been analyzed by Claustre et al. (2026). We study the consequence for driving decisions.

Critical windows and schedules. Speciation and symmetry breaking in diffusion (Biroli et al., 2024; Raya & Ambrogioni, 2023) and entropic time (Stančević et al., 2025) motivate where steps should go. A schedule that targets decision information passed a pre-registered confirmation on 436 clips not used to design it, yet its gain disappeared at matched temperature (Appendix C); this negative result motivated the protocol.

7. Discussion and Limitations

Recommendations. Report τ_{eff} for any first-order or cached sampler, and compare against the reference solver at that temperature before attributing accuracy gains to the sampler. If lower temperature is desired, set it explicitly so that it is visible and tunable.

Table 1. **Euler samplers land on the explicit-temperature curve; Heun samplers do not.** For each sampler S we report the reference solver at τ=1 (A_1), the sampler, and the reference solver at the sampler's effective temperature ($A_{\tau_{\text{eff}}}$). Flips are decision changes relative to A_1 under the same noise; the last column is the flip residual $S - A_{\tau_{\text{eff}}}$ with a 95% interval from a bootstrap over clips. 60 PhysicalAI-AV test clips × 6 seeds per model.

			ADE$_1$ (m)			Decision flips (%)		
Model	Sampler S	τ_{eff}	A_1	S	$A_{\tau_{\text{eff}}}$	S	$A_{\tau_{\text{eff}}}$	$S - A_{\tau_{\text{eff}}}$
Alpamayo-1.5	E_4	0.64	2.86	2.58	2.58	10.6	9.1	1.5 [−1.3, 5.3]
	E_{10}	0.83	2.86	2.73	2.70	4.7	4.3	0.5 [−1.2, 2.3]
	Heun-3	0.95	2.86	2.96	2.81	17.8	1.3	16.5 [9.8, 23.1]
	Heun-5	0.98	2.86	2.91	2.83	9.7	0.6	9.1 [4.3, 14.2]
Alpamayo-R1	E_4	0.61	2.47	2.10	2.15	10.8	12.9	−2.0 [−5.7, 1.2]
	E_{10}	0.84	2.47	2.28	2.30	6.1	6.1	0.0 [−2.0, 1.9]
Alpamayo-2	E_4	0.62	2.93	2.61	2.56	14.7	13.3	1.4 [−1.6, 4.9]
	E_{10}	0.83	2.93	2.78	2.76	9.7	7.7	2.0 [0.0, 4.4]

Distribution, not trajectory. Matching holds for rates averaged over clips. Under the same noise, E_4 and the reference solver at τ_{eff} still disagree on a sizable share of individual decisions, and per-clip rates differ, so the two are not interchangeable sample by sample or clip by clip.

What the test identifies. Matching spread and then comparing rates does not single out temperature. A post-hoc shrinkage of reference samples toward E_1 with the same spread also passes it. What the evidence supports is that few-step Euler is a contraction toward the conditional mean, of the size the Gaussian factor predicts, and that explicit temperature is a visible and tunable way to obtain the same rates. Heun fails because it changes decisions without contracting.

TODO: report the shrinkage arm next to $A_{\tau_{\text{eff}}}$ in Table 1.

Limitations. Decision classes are read from plans with fixed thresholds rather than from executed behavior; the reasoning trace is fixed within a clip; 60 clips do not establish equivalence at tight margins; rare decisions are untested; the positive replication comes from one model family, with Qwen-Drive as a different-family boundary case.

TODO: multi-seed language ablation; ReCogDrive and DiffusionDrive on NAVSIM.

Impact Statement

This work concerns how accelerated autonomous-driving policies are evaluated. Gains that come from a hidden temperature change can hide shifts in the decision distribution, and reporting effective temperature makes such shifts visible. We used public data and frozen public checkpoints; no vehicles were driven.

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Table 2. **Euler samplers contract toward the conditional mean; Heun does not.** Median ratio of each sample's distance to the one-step mean E_1 over the reference sample's distance, from the same noise (share of samples with ratio < 1 in parentheses), 60 clips $\times$ 6 seeds. Gaussian: the closed-form factor of Equation (1) from Alpamayo-1.5's conditional action variance (median over a separate sample of 200 clips in action space; it does not predict the per-clip ranking).

Sampler	Alpamayo-1.5	Alpamayo-R1	Alpamayo-2	Gaussian
E_4	0.68 (92%)	0.66 (93%)	0.67 (92%)	0.67
E_{10}	0.86 (90%)	0.84 (93%)	0.85 (90%)	0.86
Heun-5	0.96 (69%)	–	–	–

A. Additional Results

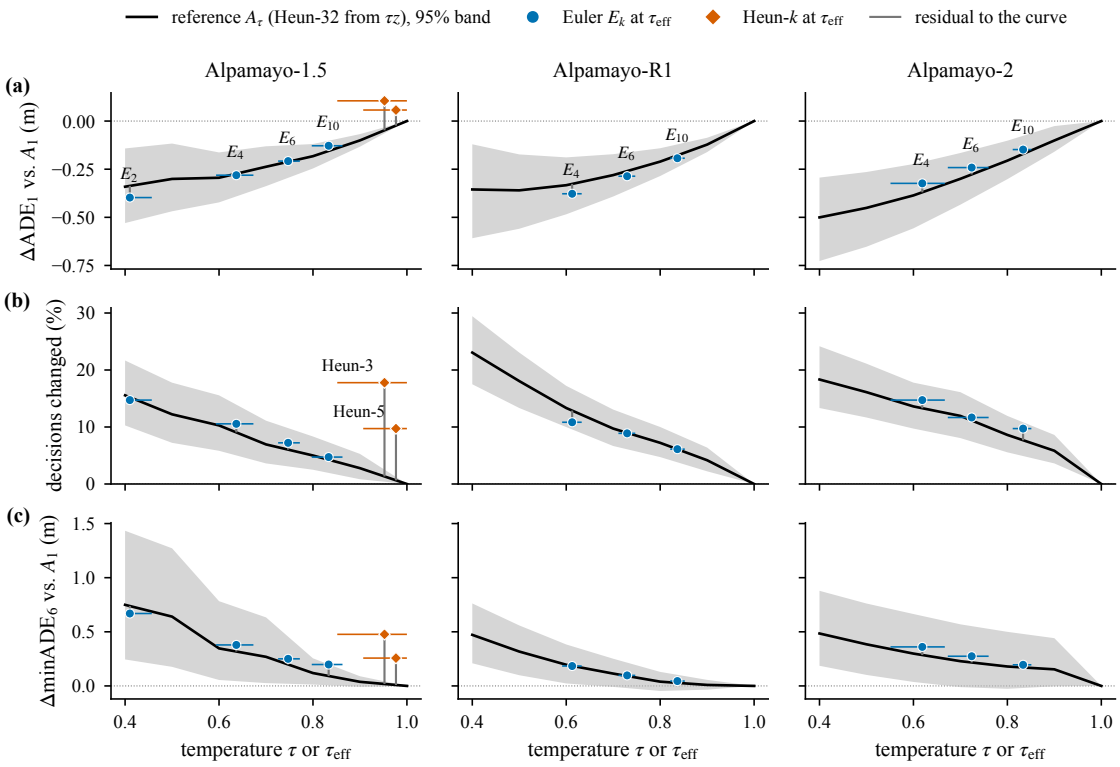


Figure 4. As Figure 3, with (c) best-of-six error (minADE_6) as a change from A_1. Euler lies slightly above the temperature curve: fewer distinct samples cost best-of-k error beyond what temperature predicts.

B. Solvable Mixture Models

TODO: K-mode experiments; note that flips under Heun and RK4 also move toward the mean in these models, so the regularity does not single out first-order samplers.

C. A Negative Result: Decision-Information Schedules

TODO: pre-registered confirmation on 436 clips (flip reduction -0.88 points, $[-1.28, -0.52]$) and its disappearance at matched temperature.

D. Manipulation: SmolVLA on LIBERO

TODO: contraction of probe samples; success insensitive to $\tau \in [0.5, 1]$; LIBERO-10 drops at very low temperature.

E. Protocols and Reproducibility

TODO: pre-registration documents, decision thresholds, seeds, data splits, compute.